Pratik Nanaso Raut

Pune, Maharashtra | +91 8600475940 | pratikraut.codex@gmail.com | https://linkedin.com/in/pratik-raut-85248b319 | https://github.com/pratik-codeX

PROFESSIONAL SUMMARY

Computer Science candidate with strong foundation in backend, cloud infrastructure, and AI-powered workflows — developing scalable platforms. Experience building backend applications using Java, Spring Boot, and PostgreSQL.

TECHNICAL SKILLS

Languages	Java, JavaScript, SQL, Python (basics), C, C++
Backend	REST APIs, Spring Boot, Spring Security, JWT, Hibernate, JDBC
Databases	MySQL, MongoDB
Tools	Git, Postman, GitHub, Maven, IntelliJ IDEA
Concepts	REST API Design, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), SDLC, Design Patterns, Problem Solving, Time Complexity Analysis, Database Optimization, RBAC, JWT Authentication, Microservice Fundamentals, API Security, Multi-Tenant Architecture, Multithreading
Exposure	AWS Fundamentals, AI/ML Fundamentals, Big Data Concepts
AI Tools	Ai

PROFESSIONAL EXPERIENCE

Software Developer Intern *Feb 2026 -- Ongoing*

SetTribe IT Solutions | Pune

- Developed and enhanced scalable backend modules for multiple client projects using Java, Spring Boot, Hibernate, and REST APIs.
- Integrated PostgreSQL database operations, validation logic, and API testing workflows using Postman.
- Collaborated in Agile sprint cycles, performing Git-based code reviews, bug fixes, feature implementation, and deployment support.

ACADEMIC PROJECTS

Customised Virtual File System (CVFS) -- In-Memory Linux-like File Manager *C, C++, Data Structures, OS Concepts*

- Simulated Linux file system operations using C++ with inode-based metadata management.
- Achieved O(1) file lookup by designing hash table-based inode indexing.
- Implemented in-memory file allocation table managing file permissions and traversal.

ProcMemInspector -- Linux Process Memory Layout Analyzer *C, Linux Internals, /proc Filesystem*

- Developed /proc filesystem parser in C to extract virtual memory layout of processes.
- Implemented process statistics monitoring by reading /proc/[pid]/maps and /proc/[pid]/status.
- Analyzed virtual address space segmentation demonstrating memory isolation.

Command Suite -- Re-implementation of Linux Core Utilities *C, Linux System Programming, POSIX System Calls*

- Re-implemented core Linux utilities using POSIX system calls including stat() and opendir().
- Handled file permission bitmask parsing, directory traversal, and process status extraction.
- Implemented full flag-based argument parsing matching standard GNU utility behavior.

TECHNICAL STRENGTHS

- Proficient in backend application development, REST API design, and database-driven systems.
- Strong debugging, query optimization, and software architecture capabilities.
- Hands-on experience building concurrent, multi-tenant, and performance-optimized Java applications.
- Practical exposure to network programming, socket communication, and concurrent server development.

EDUCATION

Master of Computer Applications (MCA) *2026 | Pursuing*

Dnyansagar Institute of Management and Research, Pune (SPPU) 7.24 CGPA **Bachelor of Computer Applications (BCA)** *2024*

C.T. Bora College, Shirur (SPPU) 73.34% **Higher Secondary Certificate (HSC)** *2021*

Maharashtra State Board 75%